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G H Raisoni International Skill Tech University, Pune

F.Y. B. Tech (TERM-I)

CAE II- WINTER 2025 (2024 Pattern)

Matrices and Differential Calculus -25UBSL1101

(Time: - 01 Hour) (Max. Marks: 20)

Instructions to the students:

i) All questions are compulsory. ii) neat diagrams must be drawn wherever necessary.
iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed
v) Assume suitable data, if necessary.

Q.No	*Question*	Marks	CO	BL
Q1.(a)	Verify Rolle's Theorem for the following function $f(x) = x^2 - 5x + 4,\ x \in [1,4]$	[5]	3	2
b)	If $y = \tan^{-1} x$, *show that* $(1-x^2)y_{n+2} + 2(n+1)xy_{n+1} + n(n+1)y_n = 0$	[5]	3	3
	OR			
Q 2. (a)	Find the nth derivative of $y = e^x \sin 4x \cos 6x$	[5]	3	2
(b)	Using Taylor's theorem express $2x^3 + 7x^2 + x - 6$ in Powers of $(x - 2)$	[5]	3	3
Q 3. (a)	If $x = u(1-v), y = uv$ *find* J *and* J' and prove that $JJ' = 1$	[5]	4	3
(b)	Verify Euler's theorem for the function $u = x^3 + y^3 + z^3 + 3xyz$	[5]	4	3
	OR			
Q. 4 (a)	If $x = r\cos\theta, y = r\sin\theta$ then evaluate $\frac{\partial(x,y)}{\partial(r,\theta)}$ and $\frac{\partial(r,\theta)}{\partial(x,y)}$.	[5]	4	3
(b)	IF $u = log(x^2 + y^2)$ find $\frac{\partial^2 u}{\partial x \partial y}$.	[5]	4	3